

Numbers and Sequences (10 th std)

Choose the Appropriate answer

1. Euclid's division lemma states that for positive integers 'a' and 'b', there exist unique integers q and r such that $a = bq + r$, where r must satisfy

- (a) $1 < r < b$ (b) $0 < r < b$ (c) $0 \leq r < b$ (d) $0 < r \leq b$

Ans: (c) $0 \leq r < b$

2. Using Euclid's division lemma, if the cube of any positive integer is divided by 9 then the possible remainders are

- (a) 0, 1, 8 (b) 1, 4, 8 (c) 0, 1, 3 (d) 1, 3, 5

Ans: (a) 0, 1, 8

3. If the H.C.F of 65 and 117 is expressible in the form of $65m - 117$, then the value of m is

- (a) 4 (b) 2 (c) 1 (d) 3

Ans: (b) 2

4. The sum of the exponents of the prime factors in the prime factorization of 1729 is

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (c) 3

5. The least number that is divisible by all the numbers from 1 to 10 (both inclusive) is

- (a) 2025 (b) 5220 (c) 5025 (d) 2520

Ans: (d) 2520

2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
2	1, 1, 3, 2, 5, 3, 7, 4, 9, 5
3	1, 1, 3, 1, 5, 3, 7, 2, 9, 5
5	1, 1, 1, 1, 5, 1, 7, 2, 3, 5
	1, 1, 1, 1, 1, 1, 7, 2, 3, 1
L.C.M. = $2^3 \times 3^2 \times 5 \times 7 = 8 \times 9 \times 5 \times 7 = 2520$	

6. $7^{4k} \equiv \underline{\hspace{1cm}} \pmod{100}$

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (a) 1

7. Given $F_1 = 1$, $F_2 = 3$ and $F_n = F_{n-1} + F_{n-2}$ then F_5 is

- (a) 3 (b) 5 (c) 8 (d) 11

Ans: (d) 11

$$F_n = F_{n-1} + F_{n-2} \Rightarrow F_5 = F_4 + F_3 = 7 + 4 = 11$$

8. The first term of an arithmetic progression is unity and the common difference is 4. Which of the following will be a term of this A.P.

- (a) 4551 (b) 10091 (c) 7881 (d) 13531

Ans: (c) 7881

$$\begin{aligned} \text{A.P. } a_n &= a_1 + (n-1)d \Rightarrow a_n = 4n - 3 \\ X &= 4n - 3 \Rightarrow n = \frac{X+3}{4} \\ 7881 + 3 &= 7884 \text{ (Divisible by 4; } n = 1971) \end{aligned}$$

9. If 6 times of 6th term of an A.P is equal to 7 times the 7th term, then the 13th term of the A.P. is

- (a) 0 (b) 6 (c) 7 (d) 13

Ans: (a) 0

10. An A.P consists of 31 terms. If its 16th term is m, then the sum of all the terms of this A.P. is

- (a) 16 m (b) 62 m (c) 31 m (d) 31 / 2 m

Ans: (c) 31 m

11. In an A.P., the first term is 1 and the common difference is 4. How many terms of the A.P must be taken for their sum to be equal to 120?

- (a) 6 (b) 7 (c) 8 (d) 9

Ans: (c) 8

12. $A = 2^{65}$ and $B = 2^{64} + 2^{63} + 2^{62} + \dots + 2^0$ which of the following is true?

- (a) B is 2^{64} more than A (b) A and B are equal
(c) B is larger than A by 1 (d) A is larger than B by 1

Ans: (d) A is larger than B by

13. The next term of the sequence $3/16, 1/8, 1/12, 1/18$ is

- (a) $1/24$ (b) $1/27$ (c) $2/3$ (d) $1/81$

Ans: (b) $1/27$

14. If the sequence $t_1, t_2, t_3 \dots$ are in A.P. then the sequence $t_6, t_{12}, t_{18} \dots$ is

- (a) A Geometric Progression
(b) An Arithmetic Progression
(c) Neither an Arithmetic Progression nor a Geometric Progression
(d) A constant sequence

Ans: (b) An Arithmetic Progression

15. The value of $(1^3 + 2^3 + 3^3 + \dots + 15^3) - (1 + 2 + 3 + \dots + 15)$ is

- (a) 14400 (b) 14200 (c) 14280 (d) 14520

Ans: (c) 14280

$$\begin{aligned} \text{The sum of the first } n \text{ natural numbers is given by the formula } \sum_{i=1}^n i &= \frac{n(n+1)}{2} \quad (n = 15) \\ \sum_{i=1}^{15} i &= \frac{15(15+1)}{2} = \frac{15 \times 16}{2} = 120 \quad \sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2 = (120)^2 = 14400 \\ 14400 - 120 &= 14280 \end{aligned}$$

16. What is the common difference of the AP 10, 7, 4...?

- (a) 3 (b) - 3 (c) 7 (d) - 7

Ans: (b) - 3

17. If the n^{th} term of an AP is $a_n = 3 + 4n$, what is the common difference?

- (a) 3 (b) 4 (c) 7 (d) 1

Ans: (b) 4

18. Find the 30th term of the AP: 10, 7, 4...

- (a) 97 (b) 77 (c) - 77 (d) -87

Ans: (c) - 77

19. If the p^{th} term of an AP is q and the q^{th} term is $(p+q)^{\text{th}}$ term?

- (a) $p + q$ (b) $p - q$ (c) $-(p + q)$ (d) 0

Ans: (c) $-(p+q)$

20. What is the sum of the first 20 terms of the AP with $a=5$ and $d=3$?

- (a) 670 (b) 620 (c) 610 (d) 600

Ans: (a) 670

21. The product of two numbers is 1729. Their HCF is 1. What is their LCM?

- (a) 1729 (b) 1 (c) 100 (d) 2025

Ans: (a) 1729

22. The value of the expression $2\sqrt{3} + 3\sqrt{3}$ is:

- (a) $5\sqrt{3}$ (b) $6\sqrt{3}$ (c) $\sqrt{18}$ (d) 5

Ans: (a) $5\sqrt{3}$

23. In an Arithmetic Progression, if $a = 28$, $d = -4$, $n = 7$, then a_n is:

- (a) 4 (b) 5 (c) 3 (d) 7

Ans: (a) 4

$$a_n = a + (n-1)d = 28 + (7-1)(-4) = 4$$

24. If $a = 10$ and $d = 10$, then first four terms will be:

- (a) 10, 30, 50, 60 (b) 10, 20, 30, 40

- (c) 10, 15, 20, 25 (d) 10, 18, 20, 30

Ans: (b) 10, 20, 30, 40

25. The first term and common difference for the A.P. 3, 1, - 1, - 3 is:

- (a) 1 and 3 (b) -1 and 3 (c) 3 and - 2 (d) 2 and 3

Ans: (c) 3 and - 2

26. 30th term of the A.P: 10, 7, 4 is

- (a) 97 (b) 77 (c) - 77 (d) - 87

Ans: (c) - 77

$$a_n = a + (n-1)d = 10 - 87 = - 77$$

27. 11th term of the A.P. $-3, -1\frac{1}{2}, 2, \dots$ is
(a) 28 (b) 22 (c) -38 (d) -48

Ans: (b) 22

$$a_n = a + (n-1)d = 3 + (11-1)(5/2) = 22$$

28. The missing terms in AP: $_, 13, _, 3$ are:
(a) 11 and 9 (b) 17 and 9 (c) 18 and 8 (d) 18 and 9

Ans: (c) 18 and 8

29. Which term of the A.P. $3, 8, 13, 18, \dots$ is 78?
(a) 12^{th} (b) 13^{th} (c) 15^{th} (d) 16^{th}

Ans: (d) 16^{th}

30. The 21st term of AP whose first two terms are -3 and 4 is:
(a) 17 (b) 137 (c) 143 (d) -143

Ans: (b) 137

31. If 17th term of an A.P. exceeds its 10th term by 7. The common difference is:
(a) 1 (b) 2 (c) 3 (d) 4

Ans: (a) 1

32. The number of multiples of 4 between 10 and 250 is:
(a) 50 (b) 40 (c) 60 (d) 30

Ans: (c) 60

33. 20th term from the last term of the A.P. $3, 8, 13, \dots, 253$ is:
(a) 147 (b) 151 (c) 154 (d) 158

Ans: (d) 158

34. The sum of the first five multiples of 3 is:
(a) 45 (b) 55 (c) 65 (d) 75

Ans: (a) 45

35. The 10th term of the AP: $5, 8, 11, 14, \dots$ is
(a) 32 (b) 35 (c) 38 (d) 185

Answer: (a) 32

36. In an AP, if $d = -4$, $n = 7$, $a_n = 4$, then a is
(a) 6 (b) 7 (c) 20 (d) 28

Ans: (d) 28

37. The list of numbers $-10, -6, -2, 2, \dots$ is
(a) an AP with $d = -16$ (b) an AP with $d = 4$
(c) an AP with $d = -4$ (d) not an AP

Ans: (b) an AP with $d = 4$

38. If the 2nd term of an AP is 13 and the 5th term is 25, then its 7th term is
(a) 30 (b) 33 (c) 37 (d) 38

Ans: (b) 33

39. Which term of the AP: 21, 42, 63, 84... is 210?
(a) 9th (b) 10th (c) 11th (d) 12th

Ans: (b) 10th

40. What is the common difference of an AP in which $a_{18} - a_{14} = 32$?
(a) 8 (b) - 8 (c) - 4 (d) 4

Ans: (a) 8

41. The famous mathematician associated with finding the sum of the first 100 natural numbers is

(a) Pythagoras (b) Newton (c) Gauss (d) Euclid

Ans: (c) Gauss

42. The sum of first 16 terms of the AP: 10, 6, 2... is
(a) -320 (b) 320 (c) -352 (d) -400

Ans: (a) - 320

43. A number when divided by 60 gives 35 as quotient and leaves 81 as remainder. What is the number?

(a) 2121 (b) 4151 (c) 2181 (d) 3171

Ans: (c) 2181

According to Euclid's Division Lemma,
 $\text{Number} = (\text{quotient} \times \text{divisor}) + \text{remainder} = 60 \times 35 + 81 = 2181$

44. For any two given positive integers a and b, there exists unique whole numbers q and r such that $a = bq + r$, where $0 \leq r < b$.

(a) True (b) False (c) Do not Determine (d) None of these

Ans: (a) True

45. When a number is divided by 3 it leaves remainder as 5. What will be the remainder when $3n + 3$ are divided by 3?

(a) 0 (b) 3 (c) 9 (d) 6

Ans: (a) 0

46. The HCF of 80 and 567 is _____

(a) 5 (b) 4 (c) 1 (d) 6

Ans: (c) 1

$80 = 1 \times 2 \times 2 \times 2 \times 2 \times 5$ $567 = 1 \times 3 \times 3 \times 3 \times 3 \times 7$.
The largest common factor between the two numbers is 1

47. A number in the form of 6^n , where n belongs to natural numbers, can never end with the digit

(a) 0 (b) 1 (c) 2 (d) 3

Ans: (a) 0

6^n can never end with 0

48.If the HCF of two numbers is 1 and the LCM is 3395. What is the other number if one of them is 97?

- (a) 61 (b) 57 (c) 43 (d) 35

Ans: (d) 35

$$(a \times b) = \text{HCF of } (a, b) \times \text{LCM of } (a, b) \quad \text{Here } a = 97, \text{ HCF is } 1 \text{ and LCM is } 3395$$
$$97 \times b = 1 \times 3395 \Rightarrow b = 3395 / 97 = 35$$

49.Which of the following is a composite number?

- (a) 2 (b) 3 (c) 9 (d) 7

Ans: (c) 9

50.Which of the following is a prime number?

- (a) 31 (b) 52 (c) 21 (d) 32

Ans: (a) 31

51.The fundamental theorem of arithmetic states that, every composite number can be factorized as product of primes and this factorization is unique.

- (a) False (b) True (c) Don't determine (d) None of these

Ans: (b) True

52.The least common multiple of 135 and 24 is _____

- (a) 90 (b) 360 (c) 240 (d) 1080

Ans: (d) 1080

$$135 = 3 \times 3 \times 3 \times 5 \times 1 \text{ and } 24 = 2 \times 2 \times 2 \times 3 \times 1$$
$$\text{Therefore, LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 5 = 1080$$

53.The highest common factor of 21 and 90 is _____

- (a) 3 (b) 2 (c) 1 (d) 4

Answer: (a) 3

$$21 = 3 \times 7 \times 1 \text{ and } 90 = 3 \times 3 \times 2 \times 5$$
$$\text{Therefore, HCF} = 3 \times 1 = 3$$

54.The LCM of two numbers is 7991 and the two numbers are 61 and 131. What will be their HCF?

- (a) 2 (b) 1 (c) 3 (d) 4

Ans: (b) 1

$$(a \times b) = \text{HCF of } (a, b) \times \text{LCM of } (a, b)$$
$$61 \times 131 = \text{HCF} \times 7991 \Rightarrow \text{HCF} = 7991 / 7991 = 1$$

55.What will be the largest number that divides 100 and 25, and leaves 3 as remainder in each case?

- (a) 7 (b) 5 (c) 1 (d) 4

Ans: (c) 1

(100-3) i.e. 97 and (25-3) i.e. 22
 Now, $97 = 97 \times 1$ and $22 = 2 \times 11$
 HCF of 97 and 22 is 1.

56. A bakery sells cookies in three boxes. The three boxes contain 60, 84 and 108 number of cookies. The baker wants to sell all the cookies in any of the three boxes. The least number of cookies that he can bake, in a day, so that he is able to sell all his cookies in any of the three boxes is _____

- (a) 5467 (b) 2243 (c) 1123 (d) 3780

Ans: (d) 3780

LCM of 60, 84 and 108 = $2 \times 2 \times 3 \times 3 \times 3 \times 5 \times 7 = 3780$

57. Two buckets contain 546 and 764 litres of water respectively. What will be maximum capacity of container which can measure the water of either buckets exact number of times?

- (a) 108 (b) 54 (c) 34 (d) 456

Ans: (a) 108

$546 = 2 \times 2 \times 3 \times 3 \times 3 \times 5$ and $764 = 2 \times 2 \times 3 \times 3 \times 3 \times 7$.
 HCF of 546 and 764 = $2^2 \times 3^3 = 108$

58. A sequence in which each term differs from its preceding term by a constant is called arithmetic progression.

- (a) True (b) False (c) Don't determine (d) None of these

Ans: (a) True

59. If we add the terms of an AP, then we get a corresponding series.

- (a) True (b) False (c) Don't determine (d) None of these

Ans: (a) True

60. Sequences which follow a definite pattern called progressions.

- (a) True (b) False (c) Don't determine (d) None of these

Ans: (a) True

61. If the first term of an AP is a and its common difference is d then the n th term is given by $T_n = a + (n - 1)d$

- (a) True (b) False (c) Don't determine (d) None of these

Ans: (a) True

62. Which term of the AP 2, 11, 20, 29..... is 290?

- (a) 32 (b) 35 (c) 30 (d) 33

Ans: (d) 33

$a + (n - 1)d = 290 = 33$

63. How many terms are there in the AP 9, 16, 23, 30.....282?

- (a) 50 (b) 45 (c) 40 (d) 42

Ans: (c) 40

$$\begin{aligned} a + (n - 1)d &= 282 \Rightarrow 9 + (n - 1)7 = 282 \\ 7n &= 280 \Rightarrow n = 40 \end{aligned}$$

64. The middle term of the AP 11, 8, 5... -79 will be?

- (a) 30 (b) -34 (c) 34 (d) -30

Ans: (b) -34

65. Which term of the AP 80, 76, 72, 68..... is the first negative term?

- (a) 22 (b) 21 (c) 23 (d) 24

Ans: (a) 22

66. What is the arithmetic mean between a^2 and b^2 ?

- (a) $12(a - b)$ (b) $12(a + b)$ (c) $12(a^2 - b^2)$ (d) $12(a^2 + b^2)$

Ans: (d) $12(a^2 + b^2)$

67. What will be the value of k so that $(2k + 1)$, $(5k - 3)$ and $(-8k + 5)$ are three consecutive terms of an AP?

- (a) 4 (b) 3 (c) 34 (d) 43

Ans: (c) 34

$$\begin{aligned} (5k - 3) - (2k + 1) &= (-8k + 5) - (5k - 3) \\ 3k - 4 &= -13k + 8 \Rightarrow 16k = 12 \Rightarrow k = 34 \end{aligned}$$

68. The sum of three numbers is 30 and their product 750. What are the three numbers?

- (a) -2, -6 and -10 (b) 2, 6, and 10
(c) 5, 10 and 15 (d) -5, -10 and -15

Ans: (c) 5, 10 and 15

69. What will be the n^{th} term of the AP 1, 5, 9, 13, 17.....?

- (a) $4n - 3$ (b) $3n - 4$ (c) $4n + 3$ (d) $3n + 4$

Ans: (a) $4n - 3$

$$\text{The } n^{\text{th}} \text{ term of the AP} = a + (n - 1)d = 1 + (n - 1)4 = 1 + 4n - 4 = 4n - 3$$

70. If the n^{th} term of the AP is $2n + 7$, then the common difference will be _____

- (a) -2 (b) 2 (c) 3 (d) -3

Ans: (b) 2

$$T_n = 2n + 7, \text{ Common difference} = T_2 - T_1 = (2 \times 2 + 7) - (2 \times 1 + 7) = 11 - 9 = 2$$

71. If the n^{th} term of the AP is $7n - 9$, then the first term is _____

- (a) 1 (b) 6 (c) 3 (d) 4

Ans: (b) 6

72. If the n^{th} term of the AP is $8n + 1$, then the 20^{th} term will be _____
 (a) 160 (b) 120 (c) 161 (d) 121

Ans: (c) 161

73. What will be the 99^{th} term from the end of the AP 500, 489, 478, 467... – 1139?
 (a) 1078 (b) 1123 (c) 12 (d) 61

Ans: (d) 61

74. If 5 times the 5^{th} term of an AP is equal to 15 times its 15^{th} term, then the value of its 20^{th} term will be _____
 (a) 0 (b) 1 (c) 2 (d) 3

Ans: (a) 0

$$\begin{aligned} 5^{\text{th}} \text{ term} &= a + (5 - 1)d = a + 4d, \quad 15^{\text{th}} \text{ term} = a + (15 - 1)d = a + 14d \\ 20d - 210d &= 15a - 5a \Rightarrow a = -19d \Rightarrow -19d + 19d = 0 \end{aligned}$$

75. If the 11^{th} term of an AP is 113 and its 13^{th} term is 111, then what will be the value of 143^{th} term?

(a) 1143 (b) 1 (c) 0 (d) 23143

Ans: (b) 1

76. If the 7^{th} term of an AP is 20 and its 11^{th} term is 40 then, what will be the common difference?

(a) 3 (b) 4 (c) 5 (d) 2

Ans: (c) 5

77. What will be the 14^{th} term of the AP 5, 8, 11, 14, 17.....?

(a) 30 (b) 41 (c) 40 (d) 44

Ans: (d) 44

78. The sum of n terms of an AP in which first term is a , common difference is d and last term is l , is $S_n = n/2 (a + l)$.

(a) True (b) False (c) Don't determine (d) None of these

Ans: (a) True

79. The sum of first 20 terms of the AP 10, 12, 14, 16, 18.... is _____

(a) 200 (b) 580 (c) 620 (d) 440

Ans: (b) 580

$$\begin{aligned} \text{We know that, } S_n &= n/2 (2a + (n - 1)d) \\ S_{20} &= 20/2 [(2(10) + (20 - 1)2)] = 10(20 + 38) = 580 \end{aligned}$$

80. The sum of $50 + 100 + 150 + 200 + \dots + 1000$ is _____

(a) 10000 (b) 14400 (c) 10500 (d) 12100

Ans: (c) 10500

81. The sum of first 16 terms of the A.P 10, 6, 2,..... is _____

(a) -320 (b) 320 (c) -352 (d) -400

Ans: (a) -320

82. If the first term of an A.P is -5 and the common difference is 2 , then the sum of the first 6 terms is _____
(a) 0 (b) 5 (c) 6 (d) 15

Ans: (a) 0

83. The 4th term from the end of the A.P $-11, -8, -5, \dots, 49$ is _____
(a) 37 (b) 40 (c) 43 (d) 58

Ans: (b) 40

84. Two A.P's have the same common difference. The first term of one of these is -1 and that of the other is -8 . Then the difference between their 4th terms is _____.
(a) -1 (b) -8 (c) 7 (d) -9

Ans: (c) 7

85. If the common difference of an A.P is 5 , then what is $a_{18} - a_{13}$?
(a) 5 (b) 20 (c) 25 (d) 30

Ans: (c) 25

86. Which term of the A.P is $21, 42, 63, 84 \dots$ is 210 ?
(a) 9th (b) 10^{th} (c) 11th (d) 12th

Ans: (b) 10th

87. If the 2nd term of an A.P is 13 and the 5th term is 25 , what is its 7th term?
(a) 30 (b) 37 (c) 33 (d) 38

Ans: (c) 33

88. The 21st term of the A.P whose first two terms are -3 and 4 is _____
(a) 17 (b) 137 (c) 143 (d) -143

Ans: (b) 143

89. If a, b, c, d, e are in A. P then the value of $a - 4b + 6c + 4d + e$ is _____.
(a) 0 (b) 1 (c) -1 (d) 2

Ans: (a) 0

90. Arithmetic mean between 8 and 16 is :
(a) 8 (b) 12 (c) 16 (d) 24

Ans: (b) 12

91. Common difference of sequence $-1, -23, -2, \dots$ is :
(a) 1 (b) -1 (c) 21 (d) -21

Ans: (d) -21

92. 5th, 8th and 11th terms of a G.P. are in :
(a) G.P. (b) A.P. (c) A.P. and G.P. both (d) H.P.

Ans: (b) A.P.

93. 7th term of sequence $2, 7, 12, \dots$ is :
(a) 34 (b) 37 (c) 32 (d) 27

Ans: (b) 37

94.If A and G are arithmetic mean and geometric mean between two positive integers then:

- (a) $A = G$ (b) $A \leq G$ (c) $A \geq G$ (d) $A + G = 0$

Ans: (c) $A \geq G$

95.Common ratio of sequence 25,45,85,..... is :

- (a) 5 (b) 41 (c) 2 (d) 21

Ans: (d) 21

96.Geometric mean of 2 and 32 is:

- (a) 2 (b) 8 (c) 64 (d) 128

Ans: (b) 8

97.If $-72, x, -27$ are in G.P. then value of x is :

- (a) ± 1 (b) 0 (c) ± 14 (d) ± 2

Ans: (c) ± 14

98.A number 3 more than 5th term of sequence 3,5,7,..... is :

- (a) 11 (b) 12 (c) 13 (d) 14

Ans: (c) 13

99.A number 2 less than 4th term of sequence 4,12,36,..... is :

- (a) 108 (b) 107 (c) 106 (d) 105

Ans: (b) 107

100. The nth term of the sequence 2, 5, 8,... is

- (a) $3n+1$ (b) $3n-1$ (c) $2n-1$ (d) $2n+1$

Ans: (c) $2n-1$